

CHAPTER 1

INTRODUCTION

1.1 Objective

The Mobile Charging Alert System (Simulation-Based) is to detect situations where a mobile device is connected to a charger but no current is being supplied, and to alert the user within a specified time limit of 30 seconds. The system aims to eliminate a common real-life problem where users forget to switch on the power supply after connecting their mobile charger, resulting in the device not being charged.

Focuses on providing an efficient and automated alert mechanism using simulation tools. The system monitors the power condition continuously and triggers alerts through a buzzer and LCD display when no current is detected for a prolonged period. By doing so, it helps users avoid inconvenience, saves time, and ensures proper device charging.

Another important objective is to design a simple, cost-effective, and user friendly system that can be easily implemented using basic electronic components in a simulated environment. The system also aims to demonstrate how automation can reduce human errors and improve everyday usability through smart monitoring techniques.

Specified Objectives with Key Functions

1 Detect No Power Condition

- Identify when charger is connected but no current is flowing
- Simulate power detection using a switch mechanism
- Continuously monitor power status

2. Time-Based Alert Mechanism

- Start a timer when no power is detected
- Trigger alert only after 30 seconds
- Prevent false alerts due to temporary power fluctuations

3. Alert System Implementation

- Activate buzzer for audible alert
- Display warning message on LCD screen
- Notify user effectively and immediately

4. Simulation-Based System Design

- Implement system using simulation tools (Tinkercad/Wokwi)
- Avoid need for real hardware during testing
- Ensure accurate behavior in virtual environment

5. Improve User Convenience

- Reduce chances of forgetting power supply
- Save time and avoid unnecessary delays
- Provide simple and easy-to-understand system output

1.2 Overview

The Mobile Charging Alert System (Simulation-Based) is a smart and practical solution designed to overcome a common problem faced by mobile users in their daily life. Often, users connect their mobile phones to a charger but forget to switch on the power supply, resulting in the device not being charged. This leads to inconvenience, wasted time, and sometimes critical situations where the phone battery drains completely when needed. The proposed system aims to eliminate this issue by providing an automated alert mechanism that detects the absence of power and informs the user within a specified time.

The system is built using an embedded system approach and is implemented in a simulation environment using tools such as Tinkercad or Wokwi. It consists of essential components such as a microcontroller (Arduino UNO), a switch to simulate power detection, a buzzer for audio alerts, and an LCD display for visual output. The switch acts

as a representation of the power supply, indicating whether current is available or not. The microcontroller continuously monitors this input and processes the data in real time.

When the system detects that the charger is connected but no power is available, it activates a timer using the internal clock function of the microcontroller. If the condition persists for more than 30 seconds, the system triggers an alert. The buzzer produces a sound to immediately notify the user, while the LCD displays a clear message such as “ALERT! Switch ON Power.” This dual alert mechanism ensures that the user is aware of the issue even if they are not directly observing the system.

Once the power supply is restored, the system automatically resets the timer and stops the alert. The LCD display updates to show a normal status message such as “Charging...”, indicating that the device is functioning properly. This automatic reset feature makes the system efficient and user-friendly, as it does not require manual intervention.

Unlike complex IoT-based systems, this project is designed entirely in a simulation environment, making it simple, cost-effective, and easy to implement. It does not require internet connectivity or additional hardware components, yet it effectively demonstrates the concept of real-time monitoring and alert systems. The project highlights how basic embedded system techniques can be used to solve everyday problems in a practical and efficient manner.

Furthermore, the system is designed using a modular approach, where different functions such as power detection, data processing, timer control, and alert generation are handled by separate modules. This improves system clarity, maintainability, and scalability. The design can be easily extended to include advanced features such as real-time current sensing, mobile notifications, or smart home integration in future implementations.

Overall, the Mobile Charging Alert System provides a reliable, efficient, and user-friendly solution for ensuring proper mobile charging. It enhances user convenience, reduces human error, and demonstrates the practical application of embedded systems and automation in solving real-world problems.

CHAPTER 2

LITERATURE SURVEY

2.1 Smart Monitoring Systems

According to **Lee and Seshia (2017)** in Introduction to Embedded Systems, smart monitoring systems are widely used in modern technology to observe and control real-time conditions in various applications. These systems continuously track parameters such as power, temperature, and device status, and generate alerts when abnormal conditions are detected.

Similarly, **Rao and Rao (2019)** highlighted that monitoring systems improve efficiency, reduce manual effort, and minimize errors in real-time applications. In the context of charging systems, monitoring plays an important role in ensuring proper device functionality. Most advanced monitoring systems use internet-based technologies; however, basic monitoring can also be implemented using simple embedded systems.

In this project, a simulation-based monitoring system is used to detect power availability in the charging circuit, demonstrating a cost-effective and simple implementation approach.

2.2 Power Detection and Alert Systems

Power detection systems are essential for identifying the presence or absence of electrical current in a circuit. According to **Patel and Shah (2020)**, these systems are widely used in industrial automation, safety systems, and home appliances to ensure proper operation.

Singh and Kaur (2021) explain that sensors are typically used to detect voltage or current flow, and a controller processes this data to generate alerts. Alerts can be delivered through various methods such as sound, light, or display systems.

In the proposed system, power detection is implemented using a simulated switch. When no current is detected, the system activates a timer, and if the condition persists for 30 seconds, an alert is generated using a buzzer and LCD display. This approach helps in avoiding false alerts caused by temporary fluctuations.

2.3 Smart Charging Systems

Smart charging systems are designed to improve charging efficiency and safety. According to **Monk (2016)** in *Programming Arduino*, such systems monitor parameters like battery level, charging status, and power conditions to ensure optimal performance.

Banzi and Shiloh (2014) also emphasized that advanced charging systems include features such as overcharging protection and automatic cutoff. However, most existing systems focus mainly on battery management and fail to address user-related issues, such as forgetting to switch on the power supply.

The proposed system addresses this gap by focusing on user behavior and ensuring that users are alerted when the charger is connected but no power is available, thereby improving usability and effectiveness.

2.4 Alert Mechanisms in Monitoring Systems

Alert mechanisms are a critical part of monitoring systems. According to Verma (2019), alerts provide immediate feedback to users when abnormal conditions occur and can be categorized into audible, visual, and digital notifications.

However, **Rao and Rao (2019)** pointed out that instant alerts can sometimes lead to false alarms due to temporary issues. To overcome this, modern systems use time-based alert mechanisms, where alerts are triggered only if the condition persists for a specific duration.

In this project, a 30-second delay-based alert system is implemented. When no power is detected, the system starts a timer, and only after confirming the persistence of the condition does it activate the buzzer and display a warning message on the LCD. This improves accuracy and system reliability.

2.5 Related Applications of Monitoring Systems

Monitoring systems are widely used across multiple domains. According to **Kushner (2011)**, embedded monitoring technologies are applied in smart homes, industrial automation, healthcare, and security systems to ensure efficient and reliable operation.

In smart homes, monitoring systems detect power failures and control appliances automatically. In industrial environments, they monitor machine performance and prevent failures. In healthcare, systems track patient conditions and provide alerts during emergencies.

The Mobile Charging Alert System is a practical application of these monitoring concepts. It focuses on detecting power absence in a charging system and notifying the user effectively. Although implemented in a simulation environment, the system demonstrates real-time monitoring, alert generation, and automation principles that can be extended to real-world applications.

CHAPTER 3

EXISTING AND PROPOSED SYSTEM

3.1 Existing System

Mobile charging is completely dependent on manual observation by the user. When a mobile device is connected to a charger, the user must ensure that the power supply is switched on and that charging is actually taking place. However, there is no system available to automatically detect whether current is flowing through the charger or not.

In many real-life situations, users plug in their mobile devices but forget to switch on the power supply. As a result, the device remains uncharged without the user realizing it. The only way to confirm charging is by manually checking the mobile screen or battery indicator, which is not always convenient.

The existing system does not provide any alert or notification when the charger is connected but no power is supplied. It also lacks any form of monitoring or automation. This makes the process inefficient and highly dependent on human attention.

Advantages of Existing System

- Simple and easy to use
- No additional hardware or system required
- Low cost (no setup needed)
- Suitable for basic usage

The existing method is straightforward and accessible to all users. Since it does not involve any additional system or technology, it is easy to use and does not require training.

3.2 Limitations of Existing System

- Requires constant manual checking
- No alert system for power failure
- High chances of user negligence

- Leads to battery not being charged
- Wastes time and causes inconvenience
- No monitoring or automation

The existing system has several limitations that reduce its effectiveness. Since there is no automatic detection, users often forget to check whether their device is charging. This results in wasted time, especially when the phone is needed urgently. Additionally, there is no feedback mechanism to inform the user about the charging status, making the system unreliable.

3.3 Proposed System

The Mobile Charging Alert System is a simulation-based solution designed to overcome the limitations of the existing system. It automatically detects whether a charger is connected but no current is being supplied and alerts the user within 30 seconds.

The system uses a microcontroller along with a switch to simulate power detection. When the charger is connected but no power is available, the system starts a timer. If the condition persists for more than 30 seconds, the system activates a buzzer and displays a warning message such as “Switch ON Power” on an LCD screen. The system continuously monitors the power condition and updates the status in real time. When power is restored, the alert is turned off, and the LCD displays a normal charging message. This ensures that the user is always aware of the charging condition without manual checking.

This project is implemented entirely in a simulation environment using tools like Tinkercad or Wokwi, making it easy to design, test, and demonstrate without requiring physical hardware.

Key Features of Proposed System

- Automatic detection of no power condition
- 30-second delay-based alert mechanism
- Buzzer for audible alert

- LCD display for visual notification
- Real-time monitoring of charging status
- Simulation-based implementation
- Simple and cost-effective design

3.4 Advantages of Proposed System

The proposed system significantly improves user convenience by eliminating the need for manual checking. It ensures that users are immediately alerted when their mobile device is not charging due to the absence of power supply. This helps in saving time and prevents unexpected battery drain.

The use of a time-based alert mechanism ensures that the system avoids false alarms caused by temporary power interruptions. The combination of buzzer and LCD display provides both audible and visual alerts, making the system effective and user-friendly.

From a design perspective, the system is simple, cost-effective, and easy to implement. Since it is simulation-based, it does not require expensive hardware and can be easily demonstrated in academic environments. The system also reduces human error and provides a reliable solution for a common real-life problem. Overall, the proposed system offers a smart, efficient, and practical approach to improving the mobile charging process through automation and monitoring.

CHAPTER 4

HARDWARE & SOFTWARE REQUIRED

4.1 Hardware Requirements

The Mobile Charging Alert System (Simulation-Based) is designed using simple and easily available electronic components, making it cost-effective and easy to implement. Since the project is developed in a simulation environment, physical hardware is not mandatory for demonstration. However, the system is based on standard embedded components that can be implemented in real hardware if required.

The core component of the system is a microcontroller such as Arduino UNO, which is responsible for processing inputs and controlling outputs. A push button or switch is used to simulate the presence or absence of power supply. This acts as a power detection mechanism in the system. The system also includes a buzzer, which provides an audible alert when no current is detected, and an LCD display (16x2), which shows warning messages such as “Switch ON Power” or normal status messages like “Charging”.

A potentiometer is used to adjust the contrast of the LCD display for better visibility. Basic connecting elements such as jumper wires and a breadboard are used to establish circuit connections in simulation. Additionally, a computer or laptop is required to run the simulation software and test the system. Overall, the hardware requirements are simple, low-cost, and easily accessible, making the system suitable for academic projects and demonstrations.

Required Hardware Components

- Arduino UNO (Microcontroller)
- Push Button / Switch (Power detection)
- Buzzer (Alert output)
- Potentiometer (for LCD contrast control)
- Breadboard (for connections)

4.2 Software Requirements

The Mobile Charging Alert System is developed and tested entirely using simulation software, making it easy to design and implement without requiring physical hardware. The software plays a crucial role in writing, testing, and executing the program that controls the system behavior.

The primary software used for coding is the Arduino IDE, which allows users to write and upload programs to the microcontroller. The programming language used is Embedded C (Arduino language), which is simple and suitable for controlling hardware components. For simulation, platforms such as Tinkercad Circuits or Wokwi are used. These tools provide a virtual environment where users can design circuits, connect components, and run simulations in real time. They also help in debugging and verifying the system before actual implementation.

A web browser such as Google Chrome or Mozilla Firefox is required to access these simulation platforms. These browsers ensure smooth operation of the simulation interface and allow users to monitor outputs such as LCD display and serial monitor. The software setup is simple, user-friendly, and does not require advanced system specifications, making it ideal for students and beginners.

Required Software Components

- Operating System: Windows / macOS / Linux
- Programming Tool: Arduino IDE
- Programming Language: Embedded C (Arduino)
- Simulation Tools: Tinkercad Circuits / Wokwi
- Web Browser: Google Chrome / Mozilla Firefox

CHAPTER 5

SYSTEM DESIGN

5.1 Block Diagram

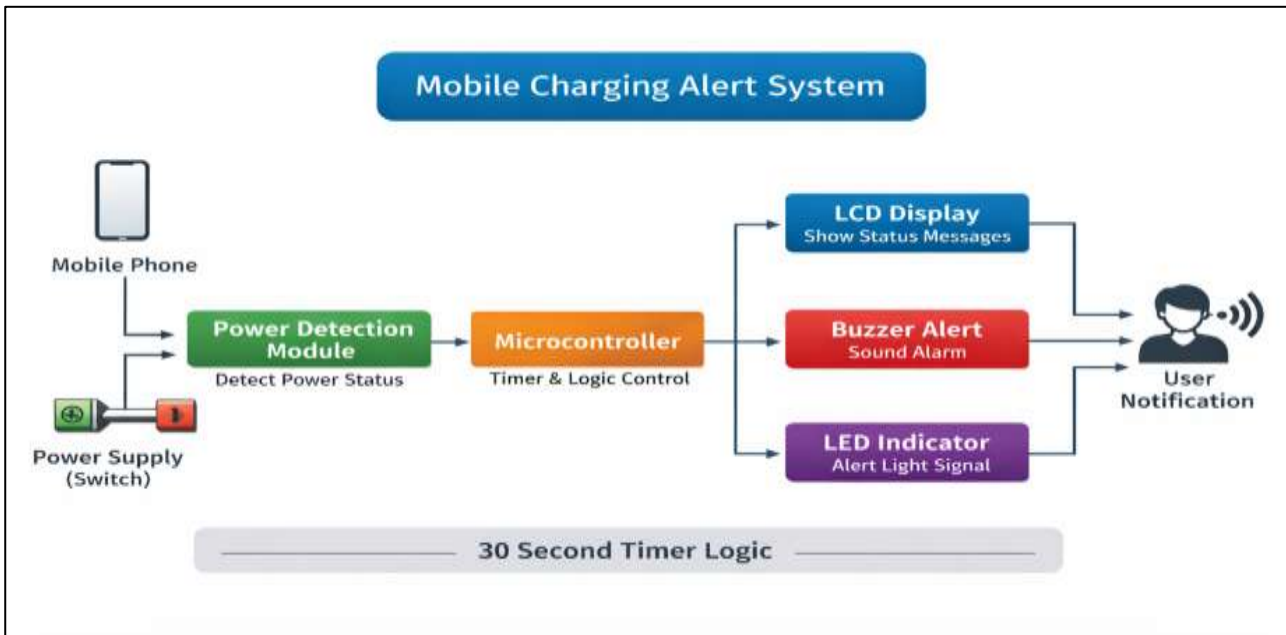


Fig 5.1 BLOCK DIAGRAM

5.2 Explanation of Block Diagram

The block diagram of the Mobile Charging Alert System provides a clear representation of the overall architecture and working mechanism of the system. It illustrates how different functional modules are interconnected and how signals flow between them to achieve continuous monitoring and alert generation. The system is designed in a structured and modular manner, where each block performs a specific task and collectively contributes to detecting the absence of power and notifying the user effectively.

At the initial stage, the system consists of a Power Detection Module, which acts as the input unit. In the simulation environment, this module is implemented using a switch or push button that represents the availability of power supply. When the switch is in the ON state, it indicates that current is flowing through the charger, and the system remains in a normal operating condition. When the switch is turned OFF, it simulates the absence of power supply, and a signal is sent to the microcontroller for further processing. This module plays a crucial role in continuously sensing the charging condition.

The Microcontroller (Arduino UNO) serves as the central processing unit of the system and acts as the brain of the entire operation. It continuously reads input signals from the power detection module and processes them using programmed logic. The microcontroller is responsible for decision-making and controlling all other modules. When it detects that no power is available, it initiates a timing process using its internal clock functions. The system does not immediately trigger an alert; instead, it monitors whether the condition persists over a defined period.

This is where the Timer Logic Module becomes essential. The timer is implemented using functions such as `millis()` within the microcontroller, which allows accurate tracking of elapsed time without interrupting system operation. The timer ensures that the system waits for a duration of 30 seconds before generating any alert. This delay mechanism is highly important because it prevents false alarms that may occur due to temporary power interruptions or loose connections. By validating the condition over time, the system improves its reliability and accuracy.

Once the timer confirms that the no-power condition has persisted continuously for 30 seconds, the system transitions to the output stage, activating the Alert Module. This module is responsible for notifying the user through both audible and visual means. The first component of this module is the Buzzer, which produces a sound signal to immediately draw the user's attention. This ensures that even if the user is not directly looking at the system, they are still alerted.

Simultaneously, the LCD Display (16x2) provides a clear visual message indicating the problem. It displays warning messages such as "ALERT! Switch ON Power," which helps the user understand the exact issue. During normal operation, the LCD shows messages like "Charging..." or "Power OK," thereby providing continuous feedback about the system status. The combination of both audio and visual alerts enhances user awareness and improves system effectiveness.

CHAPTER 6

MODULES USED

The Mobile Charging Alert System is designed using a modular approach, where the entire system is divided into different functional units called modules. Each module performs a specific task, and together they ensure smooth operation of the system. This modular design improves clarity, maintainability, and efficiency of the system.

6.1 Real-Time Data Processing

This module is responsible for continuously monitoring and processing input data from the power detection unit. It ensures that the system operates in real time and responds immediately to any change in power condition. The microcontroller reads the input signal from the power detection module and processes it using programmed logic. Based on the input, it determines whether power is present or absent and sends the result to other modules for further action.

Functions

- Continuously monitor power input
- Process data using microcontroller logic
- Identify power ON and OFF conditions
- Send processed data to alert module
- Ensure real-time system response

6.2 Power Detection

This is responsible for identifying whether current is flowing through the charging circuit. In the simulation, this is implemented using If power is restored, the microcontroller resets the timer and deactivates the alert system.

The LCD display updates to show a normal status message such as “Charging...”, indicating that the system is functioning properly. The entire system operates in a continuous loop, ensuring real-time monitoring and quick response to changes in power conditions.

Since the project is simulation-based, all these components are implemented using virtual tools, but the logic accurately represents a real-world embedded system.

Functions

- Detect presence or absence of power
- Simulate current flow using switch
- Provide input signal to microcontroller
- Enable continuous monitoring of charging status

6.3 Alert and Notification

This module is responsible for generating alerts when an abnormal condition is detected. When no power is detected, the system starts a timer. If the condition continues for 30 seconds, the module activates the alert system. The alert is provided through two outputs: a buzzer for audible notification and an LCD display for visual indication. This ensures that the user is clearly informed about the issue.

Functions

- Start timer when no power is detected
- Trigger alert after 30 seconds delay
- Activate buzzer for sound alert
- Display warning message on LCD
- Prevent false alerts using time delay

6.4 Timer Control Module

In this simulation-based system, instead of IoT communication, a Timer Control Module is used to manage time-based decision making. This module ensures that alerts are not triggered immediately but only after a specific duration. The timer is implemented using the microcontroller's internal clock (millis function). It checks whether the no-power condition persists continuously for 30 seconds before activating the alert.

Functions

- Start timer when no power is detected
- Track elapsed time continuously
- Ensure 30-second delay before alert
- Reset timer when power is restored
- Improve system reliability and accuracy

6.5 User Interface

This provides interaction between the system and the user. It ensures that the user receives clear and understandable information about the system status. This module includes an LCD display for visual output and a buzzer for audible alerts. The LCD shows messages such as “Charging...” during normal operation and “Switch ON Power” during alert conditions.

Functions

- Display system status on LCD
- Show alert messages clearly
- Provide audible alert through buzzer
- Ensure simple and use

CHAPTER 7

CONCLUSION AND FUTURE SCOPE

7.1 CONCLUSION

The Mobile Charging Alert System (Simulation-Based) successfully addresses a common real-life problem where users connect their mobile devices to a charger but forget to switch on the power supply. This often results in the device not being charged, leading to inconvenience and wasted time. The proposed system provides an effective solution by automatically detecting the absence of power and alerting the user within a specified time limit of 30 seconds.

The system continuously monitors the charging condition using a power detection mechanism and processes the input through a microcontroller. When no current is detected, a timer is activated, and after the delay period, an alert is generated using a buzzer and LCD display. This ensures that the user is immediately informed about the issue and can take corrective action.

The implementation of the system in a simulation environment demonstrates its simplicity, reliability, and cost-effectiveness. It does not require complex hardware or internet connectivity, making it suitable for academic purposes and basic real-world applications. The modular design of the system ensures smooth operation and easy understanding of each functional unit.

In conclusion, the Mobile Charging Alert System improves user convenience, reduces human error, and provides a practical and efficient solution for monitoring mobile charging conditions. It highlights how simple automation techniques can solve everyday problems effectively.

7.2 Future Scope

The Mobile Charging Alert System can be further enhanced with additional features and advanced technologies to improve its functionality and usability. In future developments, the system can be integrated with IoT technology to send real-time notifications directly to the user's mobile device, enabling remote monitoring of charging status.

The system can also be improved by using actual current or voltage sensors instead of a simulated switch, making it suitable for real-time hardware implementation. Additional features such as automatic power switching or smart plug integration can be included to automatically turn on the power supply when a charger is connected.

Further enhancements may include mobile application integration, voice alerts, and smart home compatibility, allowing the system to interact with other devices. Machine learning techniques can also be applied to analyze user behavior and provide predictive alerts or reminders. The system can be extended for multiple device monitoring, enabling users to track the charging status of several devices simultaneously. It can also be integrated into public charging stations, offices, or smart environments for better energy management and user convenience.

Overall, the future scope of the Mobile Charging Alert System is broad, and with technological advancements, it can evolve into a smart, fully automated, and intelligent charging management solution.

CHAPTER 8

APPENDIX (SOURCE CODE)

```
#include <LiquidCrystal.h>

// LCD Pin Configuration
LiquidCrystal lcd(12, 11, 10, 9, 8, 7);

// Pin Definitions
#define SENSOR_PIN 2
#define BUZZER_PIN 5

// Variables
unsigned long startTime = 0;
bool timerRunning = false;
bool alertGiven = false;

void setup() {
  pinMode(SENSOR_PIN, INPUT_PULLUP);
  pinMode(BUZZER_PIN, OUTPUT);
  lcd.begin(16, 2);
  lcd.print("System Ready");
  delay(2000);
  lcd.clear();
  Serial.begin(9600);
}

void loop() {
  int powerStatus = digitalRead(SENSOR_PIN);
  //
  NO POWER CONDITION
  if (powerStatus == HIGH) {
    if (!timerRunning) {
```

```

startTime = millis(); // Start timer

timerRunning = true;

    alertGiven = false;
    lcd.clear();
    lcd.print("No Power...");
    Serial.println("No Power Detected");
} if ((millis() - startTime >= 30000) && !alertGiven) {
    digitalWrite(BUZZER_PIN, HIGH);
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("ALERT!");
    lcd.setCursor(0, 1);
    lcd.print("Switch ON Power");
    Serial.println("ALERT: Switch ON Power!");
    alertGiven = true; // Prevent repeated alerts }
} else {
    timerRunning = false;
    alertGiven = false;
    digitalWrite(BUZZER_PIN, LOW);
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("Power OK");
    lcd.setCursor(0, 1);
    lcd.print("Charging...");
    Serial.println("Charging Normal");
    delay(1000);
}

delay(200);

```

APPENDIX B

(SCREEN SHOTS)

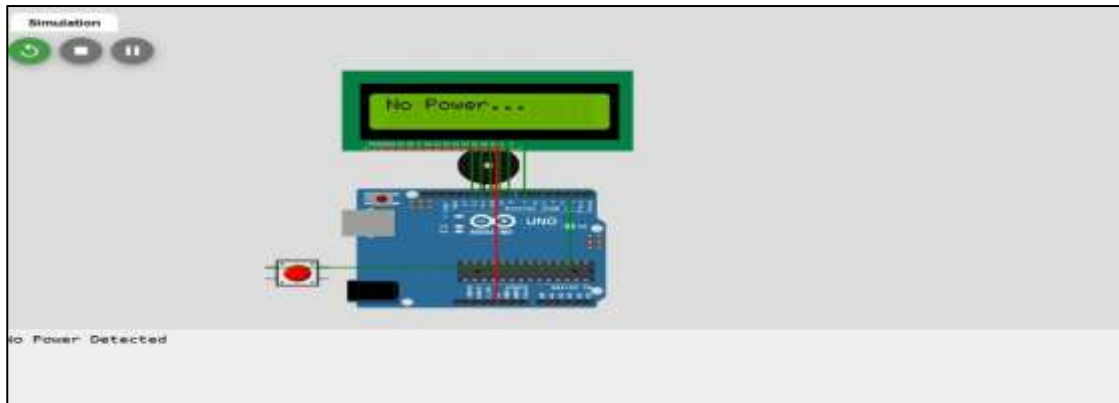


Fig B.1.1 NO POWER DETECTED

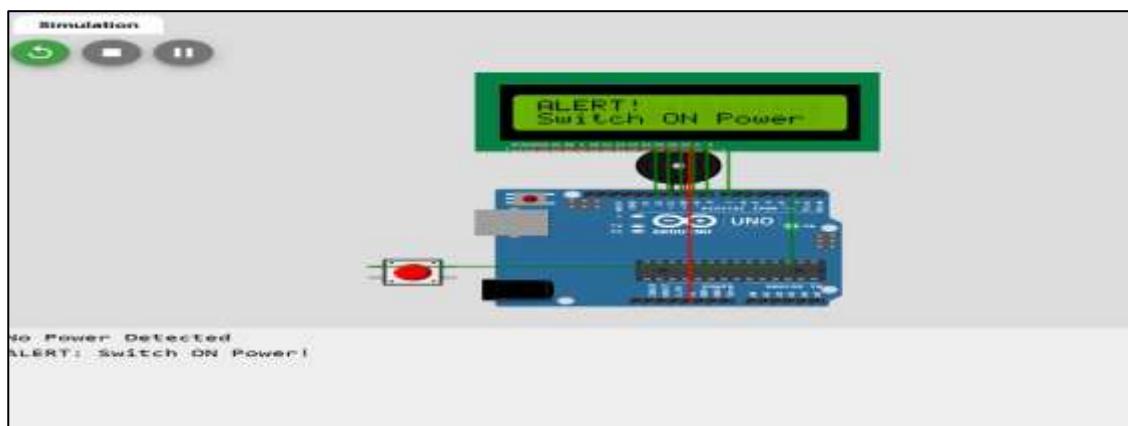


Fig B.1.2 SWITCH ON POWER

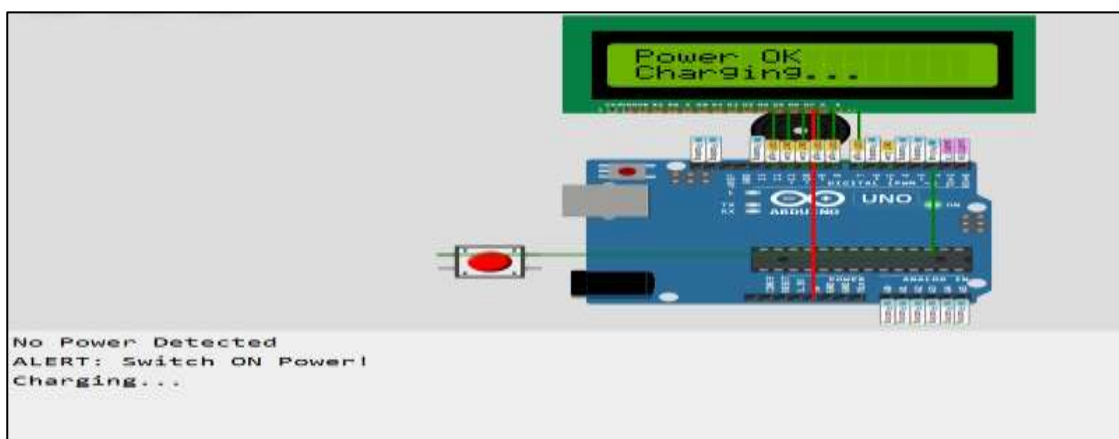


Fig B.1.3 NO POWER DETECTED

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